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Green extraction of cannabidiol from industrial hemp (*Cannabis sativa* L.) using deep eutectic solvents coupled with further enrichment and recovery by macroporous resin

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ABSTRACT

Deep eutectic solvents (DESs) are good alternatives to conventional organic solvents and ionic liquids (ILs) in the extraction and separation field. In this study, DESs were developed as the media for the green extraction of cannabidiol (CBD) from industrial hemp (*Cannabis sativa* L.). The factors influencing the extraction, including the type and concentration of DESs, solid-liquid ratio, extraction temperature, extraction time, and system pH were investigated in the single factor experiments. Then, the major factors of DESs concentration, solid-liquid ratio, extraction temperature, and extraction time were optimized by response surface methodology (RSM). Under the optimized conditions, the highest extraction yield for CBD was 12.22 mg/g. The Kamlet-Taft polarity parameters were studied to analyze the effect of polarity and hydrogen-bond ability to the extractability of DESs. The extraction kinetics was also studied and the kinetics parameters were analyzed. CBD was further enriched and recovered from the crude extract by macroporous resins, and the recovery yield and the purity reached 81.46% and 28.93%, respectively. This extraction of CBD using DESs has the advantages of high extraction yield, simple operation, low cost and good environmental friendliness, which is potential for the scale-up production.

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1. Introduction

Hemp (*Cannabis sativa* L.) is an annual and dioecious herb, which originated from Eastern and Central Asia and widely cultivated all over the world nowadays [1]. The fibers, leaves, seeds, and other by-products of hemp are widely used in the textile, materials, cosmetics, pharmaceutical, and food industries. Hemp is forbidden to be cultivated in many counties and areas because it is the raw material sources of Δ^9–tetrahydrocannabinol (THC), which is a secondary metabolite and can be used as the narcotic [2]. Therefore, to avoid the illegal use of hemp, the industrial hemp with THC content <0.2% is allowed to be cultivated in USA, Europe, China, and other countries and areas [3]. Among the various cannabinoids in cannabis, psychoactive THC and non-psychoactive CBD (its structure is shown in Fig. 1) are the most common two kinds of active compounds [4]. Both THC and CBD can be used for the medical purpose, THC is notably used in the treatment of nausea during chemotherapy and improvement the appetite for AIDS patients; and CBD is used in the treatment of treating cancer and drug-resistant epilepsy [5,6].

The traditional organic solvents, such as methanol, ethanol, chloroform, butane, and hexane were mainly used for the extraction of CBD [7]. For example, Agarwal and co-workers developed an ultrasound-assisted extraction of cannabinoids from *Cannabis sativa* L. using 80% methanol as the extractant [8]. The use of organic solvents in the extraction has the advantages of low cost, easy operation and no use of sophisticated equipment, however, their flammability, toxicity and non-biodegradability will bring potential human risks, safety problem, and environment impact [9,10]. A much greener method of supercritical fluid extraction (SFE) using CO_2 was also applied to extract CBD [7,11], satisfactory results were obtained, however the high cost also restricted the large scale production.

Deep eutectic solvents (DESs) are considered as a new class of ionic liquids (ILs) analogues, which were first reported by Abbot and his co-workers [12]. DESs are eutectic mixtures of at least one hydrogen bond acceptor (abbreviated as HBA, including choline chloride and betaine, *etc.*) and one hydrogen bond donor (abbreviated as HBD, including renewable carboxylic acids, polyols, saccharides, *etc.*) at room temperature [13,14]. The strong hydrogen bonding interaction between HBA and HBD is the most important factor for the formation of DESs [15]. DESs have the similar physicochemical properties to ILs, including

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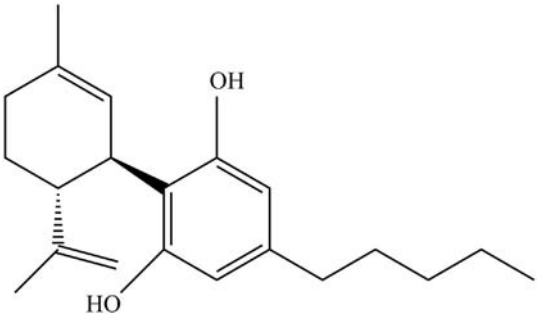


Fig. 1. The chemical structure of CBD.

high thermal and electrochemical stabilities, nonflammability, negligible volatility, and high dissolving capacity [16]. Besides, DESs have other merits including lower toxicity and higher biodegradability, lower cost, easier preparation and storage [13,17]. Thus, DESs can be used as good alternatives to conventional toxic organic solvents and ILs in the extraction and separation field, such as gas capture [18], dissolving lignocelluloses [19], separation of azeotropic mixture [20], solid-liquid extraction [21], solid-phase extraction [22], liquid-phase microextraction [23], liquid-liquid microextraction [24], and aqueous two-phase extraction [25]. The application of DESs in the solid-liquid extraction of natural products was first reported by Choi and his coworkers [26]. Then, DESs were widely used as the green media for the extraction of various bioactive compounds from plant materials, such as the extraction of polygodial from *Pseudowintera colorata* Leaves [27], artemisinin from *Artemisia annua* Leaves [28], isoflavonoids from *Dalbergia odorifera* T. Chen leaves [29], phenolic compounds *Juglans regia* L. [30], isoflavones from soy products [31], and so on. Although ILs and DESs were widely used in the extraction and purification of active compounds from natural products, the mechanism of the extraction process were rarely studied. Still some literatures had reported or summarize the probable mechanisms in the extraction process, including the driving force of thermodynamics, hydrogen bonding interaction, hydrophobic interaction, electrostatic interactions, π-π and n-π interactions [32–34].

The aim of this work is to develop DESs for the green extraction of CBD from industrial hemp. Sixteen DESs were prepared and chosen as the candidates used in this solid-liquid extraction. The factors affecting the extraction of CBD, including type and concentration of DESs, solid-liquid ratio (the solid is hemp leaves powder and the liquid is DESs aqueous solution), extraction temperature, extraction time, and system pH were studied in detail. The major extraction conditions were optimized by RSM after single factor experiments. The Kamlet-Taft polarity parameters of DESs and kinetics in the extraction process were evaluated.

2. Experimental

2.1. Plant materials

The industrial hemp leaves were provided by the Team of Annual Bast Fiber Crops, Institute of Bast Fiber Crops (IBFC), Chinese Academy of Agricultural Sciences (CAAS). The hemp was cultivated at the Bairuopu Experimental Base of IBFC in Wangcheng District, Changsha City, Hunan Province. And the hemp leaves were collected at October 2016, then dried in the air dry oven at 40 °C for 12 h till no weight loss. The dried leaves were smashed to powder and sieved to 60 meshes particle size.

2.2. Chemicals and reagents

The CBD standard (HPLC purity > 98%) was purchased from Saipu Ruisi (Beijing) Technol. Co., Ltd. (Beijing, China). The materials for the preparation of DESs were as follows: choline chloride (>98%) and betaine (>98%), and D-sorbitol (>98%) were purchased from Alladin Reagent Co., Ltd. (Shanghai, China); urea (>99.5%), oxalic acid (>98%), benzoic acid (>98%), citric acid (>98%), L-(+)-ethyl tartrate (>98%), zinc chloride (>98%), L(+)-lactic acid (>90%), glycerol (>98%), salicylic acid (>98%), succinic acid (>98%), mannitol (>98%), and acetamide (>98%) were purchased from Shanghai Titan Technol. Co., Ltd. (Shanghai, China). 2,6-dichloro-4-(2,4,6-triphenyl-N-pyridino)-phenolate (Reichardt's betaine dye 33, abbreviated as RD), 4-nitroaniline (abbreviated as NA), and *N*,*N*-diethyl-4-nitroaniline (abbreviated as DENA) were purchased from Molbase (Shanghai) Biotechnol. Co., Ltd. (Shanghai, China). The macroporous resins (Models DM-130, AB-8, D101, H103, NKA-2, and XAD-2) were purchased from Beijing Solarbio Sci. & Technol. Co., Ltd. HPLC grade methanol was purchased from TEDIA Company, Inc. (Fairfield, OH, USA). Other reagents were of analytical grade and used without further treatment. Deionized water purified by an OLABO water purification system (Model HonestDZG-303A, XinBeiXi Bio-Technology Co. Ltd., Jinan, China) was used to prepare the sample solutions.

2.3. Preparation of DESs

DESs were synthesized by the reported heating method with moderate modifications at the heating and cooling time [35]. Briefly, DESs were synthesized by simple mixing the HBAs and HBDs together at a defined molar ratio. The mixtures were constantly stirred using a magnetic stirring apparatus at 80 °C for 2–4 h until a homogeneous liquid was obtained. All the DESs were cooled down to room temperature then dried in a vacuum drying at 60 °C for 1 day. The detailed information including the crude materials, HBA/HBD molar ratios, and abbreviations for the prepared DESs were listed in Table 1. The pH values of the DESs aqueous solution were determined using a pH-meter (Model FE20, Mettler Toledo), and the results were also shown in Table 1.

2.4. Extraction of CBD using DESs

To a tube (volume of 10 mL), 0.2 g hemp leaves powder and DES aqueous solution with different concentration (50–90 wt%) at different solid-liquid ratio (1:10–1:30) were added. Then the tube was placed in a KQ-5200DE ultrasonic cleaner (Kunshan Ultrasound Co., Ltd., Jiangsu, China) at different temperatures for the ultrasonic assisted extraction of CBD. After extraction, the tube was centrifugated and the supernate was withdrawn for the analysis of CBD. The content of CBD in the crude extract was quantified by HPLC. The extraction yield of CBD in the crude extract was defined in the following Eq. (1):

$$\text{Extraction yield (mg/g)} = \frac{\text{Mass of CBD in DES extract after extraction (mg)}}{\text{Mass of hemp leaves powder (g)}} \quad (1)$$

2.5. Determination of Kamlet-Taft parameters of DESs

The Kamlet-Taft parameters were widely used to quantify the hydrogen bond donating ability (α, hydrogen-bond acidity), hydrogen bond accepting ability (β, hydrogen-bond basicity), and dipole/polarizability (π^*) of ILs and DESs using the dyes RD, NA, and DENA [36–39]. The dyes with various concentrations were mixed with DESs; both the DENA and NA concentrations were 5.0×10^{-5} mol/L, and the RD concentrations were in the range from 2.2×10^{-4} to 8.6×10^{-3} mol/L. The maximum absorption wavelength of the dye was measured using a UV–Vis spectrophotometer (Model UV-3000, Shanghai MAPADA Instruments Co., Ltd., Shanghai, China). The detailed procedures were as follows: the dyes were separately added into the DESs aqueous solution, then poured into a quartz cuvette and subjected to wavelength scanning using the UV–Vis spectrophotometer. The maximum absorption

Table 1
The Kamlet-Taft parameters and system pH were determined at 298 *K* for all the 70 wt% DESs aqueous solution.

No.	Hydrogen bond acceptors	Hydrogen bond donors	Abbreviation	Molar ratio	E_T (33) (kcal/mol)	Π^*	α	β	$\alpha \times \beta$	pH
DES-1	Choline chloride	D-sorbitol	ChCl-SB	1:1	66.804	1.244	2.216	0.493	1.093	3.86
DES-2	Choline chloride	Urea	ChCl-Urea	1:2	67.355	1.227	2.253	1.238	2.789	8.81
DES-3	Choline chloride	Oxalic acid	ChCl-OA	1:1	66.185	1.288	2.173	0.825	1.792	0.29
DES-4	Choline chloride	Benzoic acid	ChCl-BA	1:1	ND[a]	ND[a]	ND[a]	ND[a]	ND[a]	ND[a]
DES-5	Choline chloride	Citric acid	ChCl-CA	1:1	67.433	1.373	2.248	0.714	1.604	1.41
DES-6	Choline chloride	L(+)-Diethyl L-tartrate	ChCl-DT	1:1	65.880	1.770	2.118	0.840	1.779	3.38
DES-7	Choline chloride	Zinc chloride	ChCl-$ZnCl_2$	1:1	56.618	1.406	1.543	1.253	1.737	4.41
DES-8	Choline chloride	L(+)-lactic acid	ChCl-LA	1:1	67.593	1.305	2.263	1.025	3.320	1.18
DES-9	Choline chloride	Glycerol	ChCl-Gly	1:2	66.185	1.253	2.175	0.599	1.302	6.24
DES-10	Choline chloride	Salicylic acid	ChCl-SA	1:1	ND[a]	ND[a]	ND[a]	ND[a]	ND[a]	ND[a]
DES-11	Choline chloride	Succinic acid	ChCl-ScA	1:1	66.339	1.279	2.183	1.226	2.677	1.74
DES-12	Choline chloride	Mannitol	ChCl-Man	1:2	ND[a]	ND[a]	ND[a]	ND[a]	ND[a]	ND[a]
DES-13	Choline chloride	Acetamide	ChCl-AM	1:2	63.538	1.305	2.000	1.206	2.411	6.38
DES-14	Betaine	Glycerol	Bet-Gly	1:2	67.117	1.253	2.236	0.587	1.313	6.29
DES-15	Betaine	Citric acid	Bet-CA	1:1	67.275	1.671	2.216	0.012	0.031	2.69
DES-16	Betaine	Urea	Bet-Urea	1:2	65.578	1.288	2.133	1.035	2.208	8.47

[a] ND: not detectable.

wavelength was selected between 400 and 600 nm. If the maximum absorption wavelength was not between 400 and 600 nm, the first derivative of the spectrum should be selected to obtain this value. The Kamlet-Taft polarity parameters were calculated according to the following Eqs. (2)-(6) [36].

$$\nu_{max}\left(cm^{-1}\right) = 10,000/\lambda\ (nm) \tag{2}$$

$$E_T\ (33) = 28,592/\lambda_{max}\ (RD) \tag{3}$$

$$\pi* = 0.314 \times [27.52 - \nu_{max}(DENA)] \tag{4}$$

$$\alpha = 0.0649 \times E_T\ (33) - 2.03 - 0.72\pi* \tag{5}$$

$$\beta = [1.035 \times \nu_{max}(DENA) + 2.64 - \nu_{max}\ (NA)]/2.8 \tag{6}$$

where ν_{max} is the maximum experimental wave number, E_T (33) is the solvatochromic parameter in kcal/mol, λ is absorption wavelength and λ_{max} is the maximum absorption wavelength for the three dyes in the solvent.

2.6. The extraction kinetics model

The extraction of CBD using DES is always accompanied by the dissolution of a larger number of compounds into the solution. The target compounds transfer into the interface between solid and liquid from the particles of plant resources, then diffuse into the solution. Therefore, the kinetics of solid-liquid extraction of CBD was fitted according to the Fick's First Law. It was assumed that the particles of hemp was evenly distributed and the diffusion coefficient of CBD was unchanged in the solution. Eq. (7) was used to fit the kinetic model according to the reported formula derivation [40].

$$\ln\ [C_\infty/(C_\infty - C)] = kt + A \tag{7}$$

where C_∞ is the equilibrium extract concentration (mg/mL), C is the extract concentration (mg/mL) at time t, k is the pseudo first-order kinetics constant, and A is an integration constant.

2.7. Enrichment and recovery of CBD

The solid-liquid extraction using DESs can obtain the crude extract, then six kinds of macroporous resins of DM-130, AB-8, D101, H103, NKA-2, and XAD-2 (their physical properties were shown in Table S1) were used to enrich and recover CBD. The pretreatment of the resins was done according the reported procedures [41]. The crude extract was added into the glass column (10 × 500 mm) with 10 g resin (dry weight) being wet-packed in it. The crude extract flowed through the column at the flow rate of 1 BV/h. Then deionized water was added to remove the polar DESs at a flow rate of 6 BV/h, after which 95 vol% ethanol was added for the elution of the CBD at a flow rate of 2 BV/h. The collected solution was concentrated by removing the ethanol using vacuum rotary evaporation and dried in a vacuum drying oven, then CBD was recovered.

2.8. HPLC conditions

The samples were analyzed by a P230II HPLC system (Dalian Elite Analytical Instruments Co., Ltd.). This HPLC system was equipped with two P230II high-pressure constant flow pumps, a UV230II UV–Vis detector, a ZW230II column compartment, and a iChrom W5100 data processing software was used to analyze the results. A Syncronis C_{18} chromatographic column (250 × 4.6 mm, I.D. 5 μm) was used to analyze the samples. Methanol and H_2O at a volume ratio of 78:22 was used as the mobile phase. The flow rate was 0.8 mL/min with isocratic elution. The effluent was monitored at the wavelength of 220 nm. The samples were analyzed at room temperature (around 15 °C). Each sample with 10 μL was manually injected into the HPLC for analysis. The mobile phase and all the samples solution were filtered through nylon microfiltration membrane (0.45 μm) before analysis. The standard curve for analysis of UA is $Y = 1272.5 + 71{,}022X$ ($R^2 = 0.9997$), where Y is the peak area and X is the CBD concentration. The standard solutions of CBD were diluted to 0.1–1.0 mg/mL.

3. Results and discussion

3.1. Screening the optimal DES as extractant

The choice of the DES as extractant can be the most foremost thing for this solid-liquid extraction. Therefore, sixteen kinds of representative DESs were chosen as the candidates used for the extraction. Six controls for the extraction of CBD were performed using 70 wt% choline chloride (Control 1), 70 wt% L(+)-Diethyl L-tartrate (Control 2), 70 wt% mixtures of choline chloride and L(+)-Diethyl L-tartrate (Control 3) with the molar ratio of 1:1, 70 wt% methanol (Control 4), 70 wt % ethanol (Control 5), and 70 wt% ethyl acetate (Control 6). The results of the extraction yield using different DESs were shown in Fig. 2, it can be seen that the maximum extraction yield was obtained using the ChCl-DT as the extractant among the DESs. The ChCl-DT had better extractability than the organic solvents of ethanol and ethyl acetate, and a little poorer extractability than methanol. The biggest merit for DESs is that they are much environmentally friendly than all the commonly used organic solvents. More than half of the DESs used as extractant

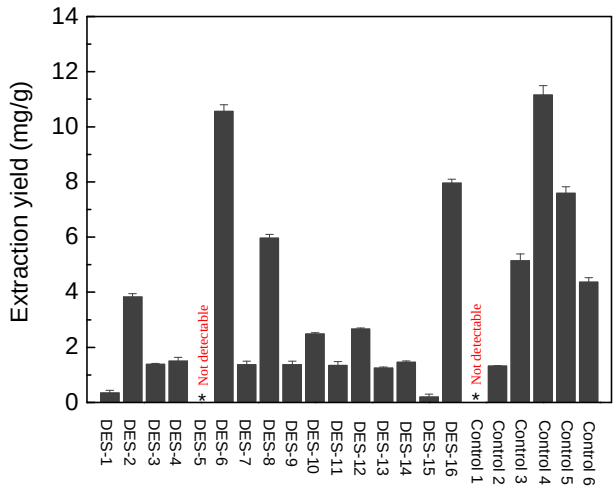


Fig. 2. Extraction of CBD from the hemp leaves using different DESs. The extraction conditions were as follow: 70 wt% extractant concentration, 30 °C extraction temperature, 40 min extraction time, and 1:20 solid-liquid ratio.

had the lower extraction yields (below 2.0 mg/g), even similar to the Control 2 of only using L(+)-Diethyl L-tartrate as the extractant. The CBD content for Control 1 can't be detectable, which indicated that various interactions between DESs and target compounds played a great role in this extraction. It was reported that the probable hydrogen-bond interaction, hydrophobic interactions, electrostatic interactions, van der Waals forces can affect the extraction ability of DESs [14].

Among the intrinsic characters of DESs, polarity is one the most important properties because it is the key indicator of the DESs' dissolving capacity, and most solvents were classified according to their dissolving ability to polar or charged species [42,43]. Therefore, to make clear the probable factor influencing the extractability of DESs, the Kamlet-Taft polarity parameters of the DESs were evaluated. The results for the α (hydrogen bond donating ability), β (hydrogen bond accepting ability), and π^* (dipole/polariz ability) were shown in Table 1. The polarity parameters were measured for each sample at the 70 wt% DES concentration used for the extraction instead of the pure DESs. Because the 70 wt% aqueous solutions for ChCl-BA, ChCl-SA, ChCl-Man were not completely dissolvable at room temperature, so their polarity parameters can't be determined and their system pH values were not determined either. It can be seen in Table 1 that majority DESs have the polarity (Π^*) among 1.2 and 1.4, and the ChCl-DT have the maximum Π^* value. The α is the hydrogen bond donating ability, β is hydrogen bond accepting ability, and it was reported that the $\alpha \times \beta$ value can represent the hydrogen bond ability [44]. CBD has two large alkyl groups in its structure (seeing Fig. 1), which makes CBD being not soluble in water. Moreover, CBD also has two phenolic hydroxyl groups, and thus the hydrogen-bond interaction can be occurred between CBD and DESs, which makes the DESs become good solvent for dissolving CBD. The DES ChCl-LA has the maximum hydrogen-bond ability (maximum $\alpha \times \beta$ value), but its extraction yield wasn't highest. To sun up these results, it can be speculated that maybe the polarity can be an important factor influencing the extraction ability for ChCl-DT, and the hydrogen bond ability is also a non-negligible factor influencing the extraction. Overall, it is a complex process for the extraction using DESs and there are many other factors can influence the extraction ability of DESs.

3.2. Single factor experiments

There are many factors, including the DES concentration, solid-liquid ratio, extraction temperature, extraction time, and pH, can affect the solid-liquid extraction. Therefore, it is necessary to investigate these factors to obtain the optimal extraction conditions. When the effect of one factor was studied, proper gradient range for it was chosen, and the remaining operating conditions were kept constant for other factors.

3.2.1. Effect of DES concentration

The DES concentration can be a major factor influencing the extraction. The addition of water is an efficient method to reduce the viscosity and adjust the solubility of compounds in DESs [45]. However, because water has strong polarity, high water content can break the balanced interaction between HBD, HBA, and the target compounds [46]. It was reported that when the water content was up to 50%, the molecules behaved like as in an aqueous solution [47]. Thus, the DES concentration range of 50–90 wt% was investigated. The results in Fig. 3a showed that the extraction yield sharply increased at the DES concentration from 50 wt% to 70 wt%, then no obvious increase with the further increase from 70 wt% to 90 wt%. The reasons for explaining this was that high water content can increase the polarity of the extraction media, and the hydrogen-bond interaction equilibrium between HBA, HBD and target compounds was destroyed, resulting in the negative impact on the extraction of CBD [48]. The results of Kamlet-Taft polarity parameters for the system of 50–90 wt% DES were provided in Table S2, it can be seen that the hydrogen-bond ability ($\alpha \times \beta$ value) roughly increased with increase of water content, which proved that high water content increased the hydrogen-bond interaction in the solution system. But high water content destroyed the hydrogen-bond interaction between the DES and target compounds resulting in the decrease of extraction yield. Furthermore, although the high DES content is beneficial for the extraction, the system viscosity simultaneously increased, which is adverse to the separation of target compounds and the large scale application. Therefore, the DES concentration of 70 wt% was chosen for further studies.

3.2.2. Effect of solid-liquid ratio

The solid-liquid ratio can be another important factor affecting the solid-liquid extraction. The insufficient solvent can lead to inadequate extraction, while the excess solvent will bring about the waste of extractant, even is adverse to the extraction. The effect of the solid-liquid ratio in the range of 1:10, 1:15, 1:20, 1:25, and 1:30 was studied. The results in Fig. 3b showed that the extraction yield varied little (Student's *t*-test, $p > 0.05$) at the solid-liquid ratios of 1:10, 1:15, and 1:20; then increased with further decrease of solid-liquid ratios to 1:25 and 1:30, but also the extraction yields varied little (Student's *t*-test, $p > 0.05$) at that conditions. The similar tendency was observed at the extraction of cynaropicrin from *Cynara cardunculus* L. leaves using DES [9]. The low solid-liquid ratio can reduce the system viscosity and promote the diffusion process of target compounds into the solution [49], which is beneficial for the extraction. Therefore, the solid-liquid ratio of 1:25 was chosen for further studies.

3.2.3. Effect of extraction temperature

Temperature is also a non-negligible factor the solid-liquid extraction. The elevation of temperature can accelerate the mass transfer and diffusion and reduce the solvent viscosity, which is beneficial for the extraction and separation. However, the higher temperature will also increase the energy consumption and break the stability of some heat-sensitive compounds. Because the 70 wt% DES aqueous solution for ChCl-BA, ChCl-SA, ChCl-Man are not completely soluble at low temperature, therefore, the effect of the temperature in the range of 40–80 °C was studied in this study. Moreover, it was reported that CBD can be converted from cannabidiolic acid (CBDA) by light or heating [50], so one more sample were prepared without heating. It was found that CBD hardly can be detected in the sample without heating. The results in Fig. 3c showed that the extraction yield increased with the increase of temperature, however, slight increase was observed at the temperature from 50 to 80 °C. Higher temperatures can provide more heat and soften the plant tissue, which make the target compounds more easily diffuse into the extraction media and enhance the extraction yield

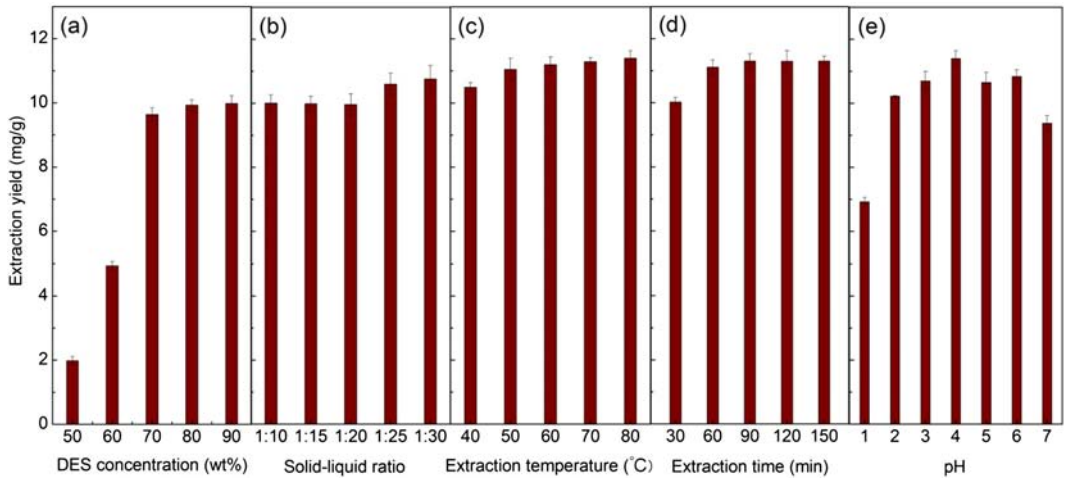


Fig. 3. Effect of the single factor to the extraction yield of CBD using ChCl-DT as extractant. (a) DES concentration was in the range of 50–90 wt%, the other conditions were as follow: 1:20 solid-liquid ratio, 40 °C extraction temperature, 60 min extraction time, and the pH was not adjusted. (b) solid-liquid ratio was in the range of 1:10–1:30, the other conditions were as follow: 70 wt% DES concentration, 40 °C extraction temperature, 60 min extraction time, and the pH was not adjusted. (c) extraction temperature was in the range of 40–80 °C, the other conditions were as follow: 70 wt% DES concentration, 1:25 solid-liquid ratio, 60 min extraction time, and the pH was not adjusted. (d) extraction time was in the range of 30–150 min, the other conditions were as follow: 70 wt% DES concentration, 1:25 solid-liquid ratio, 50 °C extraction temperature, and the pH was not adjusted. (e) pH was in the range of 1–7, the other conditions were as follow: 70 wt% DES concentration, 1:25 solid-liquid ratio, 50 °C extraction temperature, and 60 min extraction time.

[49]. However, the excessive temperature will probably cause the decomposition of the natural ingredients [48]. Therefore, to synthetically consider the extraction yield and the cost, the extraction temperature of 50 °C was chosen for further studies.

3.2.4. Effect of extraction time

Generally speaking, the extraction is expected to be finished in a short time, the long extraction time will increase the cost, even is adverse to the extraction. The effect of the extraction time in the range of 30–150 min was investigated. The results in Fig. 3d showed that the extraction yield increased from 30 to 60 min, then varied little (Student's *t*-test, $p > 0.05$) at the longer extraction time. Similar tendencies were observed at the extraction of phenolic acids from *Prunella vulgaris* L. [51] and isoflavonoids from *Dalbergia odorifera* T. Chen leaves [29], respectively using DESs as the extraction media. Therefore, the extraction time of 60 min was chosen for further studies.

3.2.5. Effect of pH

There are two phenolic hydroxyl groups in CBD (sing the Fig. 1), and thus it can be deduced that CBD is a weakly acidic compound. The effect of the pH in the range of 1–7 was studied by adjusting the pH using B–R (Britton–Robinson) buffer. The results in Fig. 3e showed that the extraction yield increased at the pH range of 1–4, then decreased with the increase of pH. The extraction system can result in a system pH around 3.38 without adjusting the pH (Seeing Table 1), and thus the extraction was done without adjusting the system pH.

3.3. Optimization of extraction by RSM

The single factor experiments can't explain mutual interactions between experimental variables. Therefore, RSM based on Box–Behnken design (BBD) was developed to investigate the multiple factors and their mutual interactions, including extraction temperature (40, 50, and 60 °C), extraction time (30, 60, and 90 min), DES concentration (60, 70, and 80 wt%), and solid-liquid ratio (1:20, 1:25, and 1:30). Because the objective of the work was to obtain the maximum extraction yield, so the larger-the-better (LTB) responses were adopted for the optimization process. The experimental results for BBD and the analysis of variance (ANOVA) for the quadratic response surface model were shown in Table 2. The ANOVA showed a *P*-value was <0.0001, which indicated regression model was significant. The Model *F*-value of 484.50 implied that the model was significant. There is a <0.01% probability that a "Model *F*-Value" occurs due to noise. The model terms can be considered to be significant when the *P*-values are <0.05. The model terms of *A*, *B*, *C*, *D*, *AC*, A^2, B^2, C^2, and D^2 were considered as significant terms according to this principle above.

The "Lack of Fit *F*-value" of 0.6851 implied the "Lack of Fit" is not significant relative to the pure error and that there is a 99.79% chance that a "Lack of Fit *F*-value" this large could occur due to noise. The regression model was generated by the software and present in Eq. (8):

$$\begin{aligned} Y = {} & 12.26 - 0.050 \times A - 0.021 \times B - 0.32 \times C - 0.095 \times D + 2.625 \\ & \times 10^{-4} \times A \times B + 0.039 \times A \times C - 0.011 \times A \times D + 0.022 \times B \\ & \times C - 5.775 \times 10^{-3} \times B \times D + 2.625 \times 10^{-4} \times C \\ & \times D - 0.13 \times A^2 - 0.075 \times B^2 - 0.93 \times C^2 - 0.35 \\ & \times D^2 \quad \left(R^2 = 0.9979, \text{the Adj and Pred } R^2 \text{ were 0.9959 and 0.9912, respectively}\right) \end{aligned} \tag{8}$$

where *Y* is the extraction yield (mg/g), *A* is the extraction temperature (°C), B is the extraction time (min), *C* is the DES concentration (wt%), and *D* is the solid-liquid ratio. The coefficient of determination (R^2) was 0.9979, implying that this quadratic model was quite suitable for the description of the response of the experiment. It can be deduced from the coefficient (absolute values) of four single factors in Eq. (8) that the factors influencing the CBD extraction followed the order as below: DES concentration > solid-liquid ratio > extraction temperature > extraction time. The parameter estimates revolved in the optimization model for BBD were listed in Table S3. The response plots for the effect of the independent variables on the average extraction yield of CBD were shown in Fig. 4(a-f). The variation tendency for mutual interaction between each two factors can be analyzed from the 3D response surfaces.

On the basis of the quadratic model, the calculated optimal conditions for the extraction of CBD are 47.88 °C extraction temperature, 55.23 min extraction time, 68.20 wt% DES concentration, and 1:24.35 solid-liquid ratio. The predicted value of extraction yield was 12.30 mg/g at the optimal conditions above. The experiments were done at the optimal conditions (48 °C extraction temperature, 55 min extraction time, 68 wt% DES concentration, and 1:24 solid-liquid ratio), the obtained extraction yield was 12.22 mg/g after calculation the mean value of triplicate runs. The experimental value was close to

Table 2
Experimental results for the four-factor/three-level Box–Behnken design and analysis of variance (ANOVA) for the quadratic response surface model.

Run	(A) Extraction temperature (°C)	(B) Extraction time (min)	(C) DES concentration (wt%)	(D) Solid-liquid ratio	Extraction yield (mg/g)
1	60	60	70	1:30	11.639
2	60	30	70	1:25	12.019
3	50	30	60	1:25	11.639
4	40	60	70	1:20	11.907
5	50	60	60	1:30	11.169
6	50	90	60	1:25	11.527
7	50	60	70	1:25	12.243
8	50	60	70	1:25	12.287
9	50	30	70	1:30	11.773
10	60	60	80	1:25	10.834
11	60	90	70	1:25	12.063
12	50	30	70	1:20	11.952
13	40	60	60	1:25	11.661
14	50	60	80	1:20	10.767
15	60	90	70	1:25	11.997
16	50	60	60	1:20	11.371
17	50	60	70	1:25	12.219
18	50	90	80	1:25	10.924
19	40	30	70	1:25	12.086
20	50	60	70	1:25	12.243
21	50	90	70	1:30	11.728
22	50	60	80	1:30	10.567
23	50	30	80	1:25	10.947
24	50	60	70	1:25	12.309
25	50	90	70	1:20	11.930
26	40	60	70	1:30	11.750
27	40	60	80	1:25	10.902
28	60	60	70	1:20	11.840
29	60	60	60	1:25	11.438

Source	Sum of squares	Degree of freedom	Mean square	*F*-Value	*P*-value Prob > *F*
Model	7.34	14	0.52	484.50	<0.0001
A	0.030	1	0.030	27.79	0.0001
B	5.074×10^{-3}	1	5.074×10^{-3}	4.69	0.0481
C	1.24	1	1.24	1150.70	<0.0001
D	0.11	1	0.11	100.16	<0.0001
AB	2.074×10^{-7}	1	2.074×10^{-7}	2.548×10^{-4}	0.9578
AC	6.037×10^{-3}	1	6.037×10^{-3}	5.58	0.0332
AD	5.096×10^{-4}	1	5.096×10^{-4}	0.47	0.5037
BC	1.945×10^{-3}	1	1.945×10^{-3}	1.80	0.2013
BD	1.334×10^{-4}	1	1.334×10^{-4}	0.12	0.7307
CD	2.756×10^{-7}	1	2.756×10^{-7}	2.548×10^{-4}	0.9875
A^2	0.11	1	0.11	102.75	<0.0001
B^2	0.036	1	0.036	33.60	<0.0001
C^2	5.60	1	5.60	5180.86	<0.0001
D^2	0.79	1	0.79	729.38	<0.0001
Residual	0.015	14	1.082×10^{-3}		
Lack of fit	9.811×10^{-3}	10	9.811×10^{-3}	0.74	0.6851
Pure error	5.335×10^{-3}	4	1.334×10^{-3}		
Cor total	7.35	28			

the predicted value, which proved that this model is adequate for predicting the expected optimization. To compare our studies with the similar reports about the extraction of cannabinoids using hydrophobic DESs [50], they studied four compounds [tetrahydrocannabinolic acid (THCA), THC, CBDA, and CBD], and the effects of DES composition and comparison to organic solvents were studied, however, the extraction yield for CBD was not very high (about 5 mg/g) in their work. In our studies, the industrial hemp with THC content <0.2% was chosen as the raw materials, and the hydrophilic DESs were chosen as the extractants, the obtained maximum extraction yield was 12.22 mg/g for CBD under the optimized extraction conditions.

3.4. Kinetics model for this solid-liquid extraction

The extraction yields were determined at different extraction temperatures and extraction time, and the data was provided in Table S4. The data was fitted to obtain the Fig. 5, which indicated the relationship between $\ln[C_\infty/(C_\infty - C)]$ and extraction time (t) at different temperatures. The regression equations, the related correlation coefficients (r^2) and calculated C_∞ were provided in the Table S5, all the correlation coefficients were >0.99 indicating the model was well fitted. This kinetics model provided some useful some information about of kinetics parameters and the effect of temperature in the extraction process. The obtained regression data can make it easy to understand the initial state for the extraction process. The temperature can accelerate the initial extraction rate and some other extraction parameters, but it can't significantly change final equilibrium concentration and extraction yield [40]. This extraction model can provide some guidance for the practical production.

3.5. Enrichment and recovery of CBD from the crude extract

The using various resins is the most commonly used method for the enrichment and recovery of the target bioactive compounds from the DES extract in solid-liquid extraction [52–56]. In order to enrich and recover CBD from the crude extract, six types of macroporous resins were

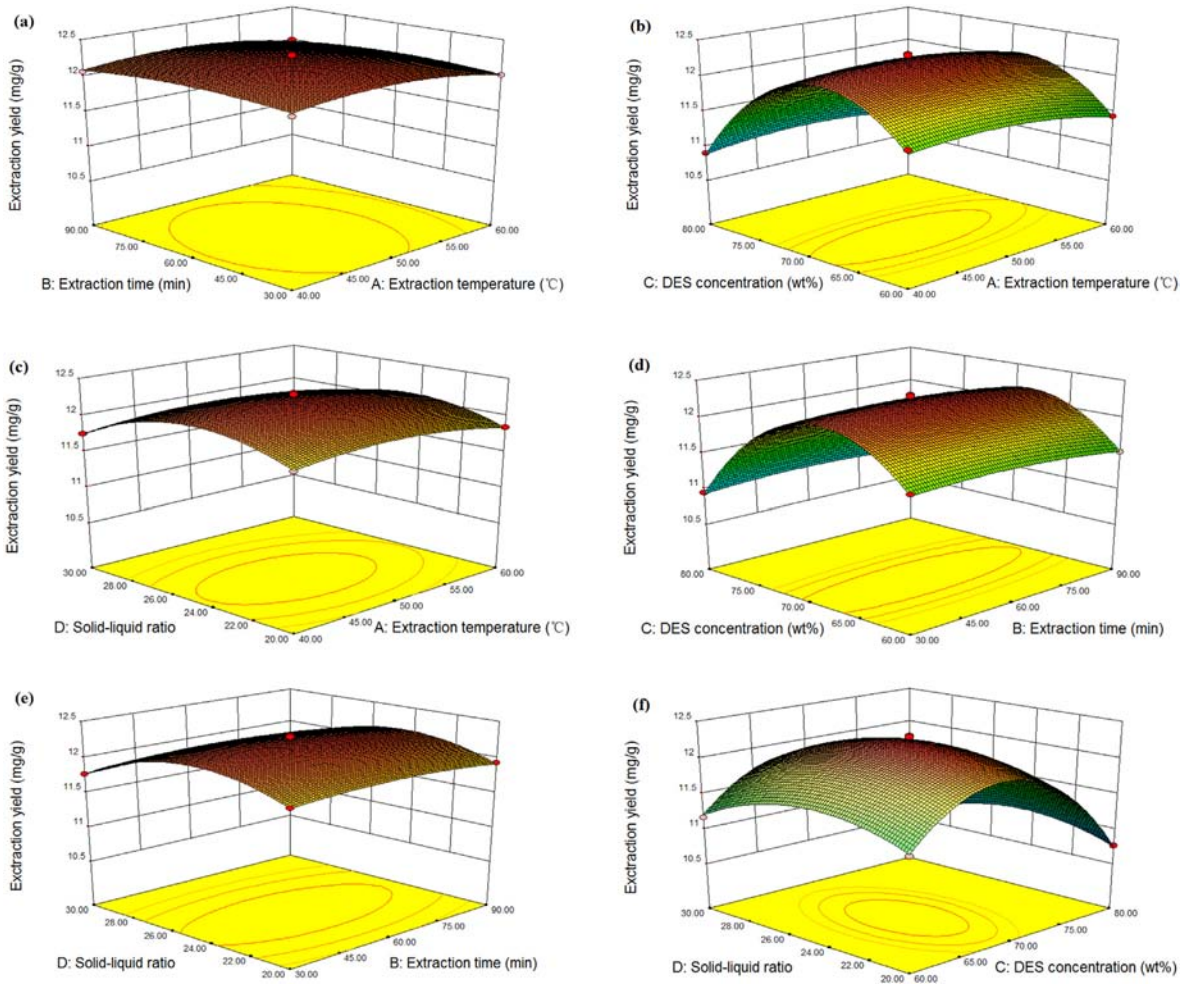


Fig. 4. Response surface plots for the extraction of CBD from hemp using ChCl-DT as extractant indicating the interaction between (a) extraction temperature and extraction time, (b) extraction temperature and DES concentration; (c) extraction temperature and solid-liquid ratio; (d) extraction time and DES concentration; (e) extraction time and solid-liquid ratio; (f) DES concentration and solid-liquid ratio.

used for the adsorption the CBD with the polar ingredients of DESs being eluted by water. Then CBD adsorbed in the resins can be eluted by ethanol. The recovery yields of different macroporous resins were as follows: DM-130, 81.46%; AB-8, 72.34%; D101, 71.28%; H103, 42.34%; NKA-2, 59.13%; and XAD-2, 61.22%. The macroporous resins DM-130 was chosen because it has the optimal adsorbing capacity for CBD. The purity of CBD obtained after the enrichment and recovery by DM-130 was determined, and the purity of 28.93% was obtained in the dried CBD sample.

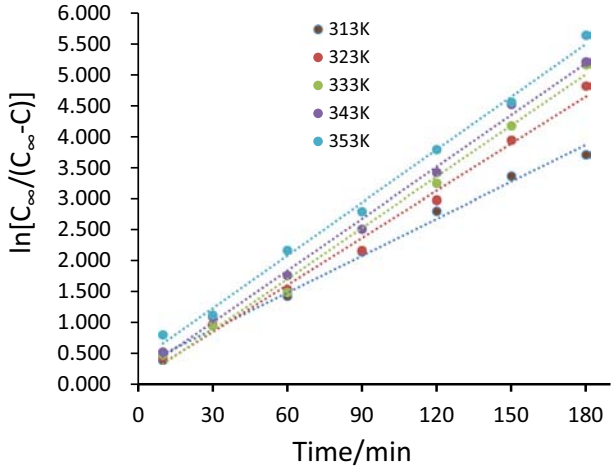


Fig. 5. Relationship between $\ln[C_\infty/(C_\infty - C)]$ and extraction time (t) at different temperatures.

4. Conclusions

DESs were developed as the solution media for the extraction of CBD from industrial hemp (*Cannabis sativa* L.) leaves in this study. The DES ChCl-DT was screened as the optimal extractant. The Kamlet-Taft polarity parameters for DESs were measured and the probable influence of polarity and hydrogen-bond interaction were analyzed. The single factor influencing the extraction was studied, then the major factors were optimized by RSM. Under the optimized conditions of 48 °C extraction temperature, 55 min extraction time, 68 wt% DES concentration, and 1:24 solid-liquid ratio, the practical extraction yield was 12.22 mg/g. The kinetics model for this solid-liquid extraction was also studied by the Fick's Law, this model was well fitted and it can provide some guidance for the practical production. After the extraction, CBD was enriched and recovered by macroporous resins, the recovery yield and purity of obtained CBD were 81.46% and 28.93%, respectively using DM-130 resins. This extraction of CBD using DESs is much greener, simpler, and more efficient, which is potential for the application in scale-up production.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2019.110957.

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